Jithendra Puppala

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Summary

MS CS at NYU with 2+ years as a Data Scientist at Jio Platforms, shipping production ML to 150K+ IoT devices ($10M+ annual savings). CV/ML intern at Applied Materials, automating wafer-inspection recipe tuning with Bayesian optimization. Leading RAG and workflow automation at NYU. Kaggle: 1st in computer vision (50+ teams), 3rd in tabular ML (100+ teams).

Education

New York University — New York, NY
Master of Science in Computer Science | GPA: 3.83 — Expected: 05/2027
- Coursework: Deep Learning (Prof. **Yann LeCun**), Machine Learning, Computer Vision, Big Data
- **Work Authorization:** US work-eligible via CPT (internship) and STEM OPT (3 years post-graduation)

National Institute of Technology Karnataka — Surathkal, India
Bachelor of Technology in Computer Science and Engineering — 05/2023
- Coursework: Data Structures and Algorithms, Digital Image Processing, Probability & Statistics, Database Systems

Skills

- **Languages & Core ML**: Python, SQL, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face, LangChain
- **Deep Learning**: Vision Transformers (ViT, Swin, EVA02), CNNs (ConvNeXt), LLMs/Transformers, CUDA, Mixed Precision, Flash Attention, Natural Language Processing
- **Computer Vision**: OpenCV, YOLO, Object Detection & Tracking, Image Classification, Multi-View Fusion
- **MLOps & Cloud**: Docker, FastAPI, REST APIs, AWS (EC2, S3), GCP, Azure, PySpark, MLflow, CI/CD, Git, Linux
- **Analytics**: A/B Testing, Hypothesis Testing, Feature Engineering, Causal Inference, Data Wrangling, Pandas, NumPy

Work Experience

Applied Materials — Santa Clara, CA
Computer Vision & Machine Learning Intern — 06/2026 – 08/2026
- Reduced **missed defects by 67%** (held-out coverage 95.1% to 98.4%) at a fixed false-alarm budget, or **false alarms 5.4×** at matched recall, via **bilevel Bayesian optimization** of wafer-inspection recipes
- Discovered an image feature separating defects from nuisance at **AUC 0.956** vs. the engine's best 0.693 (200/200 bootstrap wins); showed Bayesian search recovers the production recipe in a median of **13 evaluations vs. 103**
- Caught a coverage metric reporting a **false 100%** (85% of on-defect detections were on the wrong die), fixed it, and shipped **1,702 automated tests** and an offline React + FastAPI dashboard

New York University – Office of Faculty Affairs — New York, NY
Lead Developer – Data Science & AI — 01/2026 – Present
- Built **RAG** ticketing prototype (LangChain, ChromaDB, sentence-transformers) for faculty document Q&A at **94%** retrieval accuracy, scaling to auto-resolve **900+ annual requests** across **15+ administrative workflows**
- Developing automated **workflow translation layer** using Python, converting legacy manual processes into structured digital pipelines with **real-time status dashboards** for **300+** faculty

Jio Platforms Limited — Bengaluru, India
Data Scientist — 06/2023 – 07/2025
- Owned end-to-end ML classification pipeline serving real-time predictions to **150K+ IoT devices** via FastAPI microservices, improving **precision from 68% to 91%** and **recall from 76% to 95%**
- Built ARIMA-based anomaly detection that **reduced alert latency from 12 hours to 4 hours** (67% faster), improving field response success rates by **28%** and driving **$10M+** in annual savings
- Re-architected ML retraining infrastructure: model retraining from **18 hours to 2 hours** (89% reduction), data processing (ETL) from **3 hours to 7 minutes** (96% reduction)

Projects

Multi-View Vision Transformers for Geospatial Prediction *(1st Place, Kaggle, 50+ Teams)* — 11/2025 – 01/2026
- Developed multi-view attention architecture fusing 4 street-view images for 50-state classification + GPS regression, fine-tuning 7 vision backbones (ViT-CLIP, EVA02, Swin, BEiT, ConvNeXt)
- Achieved **95.7% score** via weighted ensemble with ablation-informed model selection and Test-Time Augmentation (TTA)
- Engineered GPU-optimized pipeline with **tensor caching** and **on-GPU** MixUp & CutMix, cutting A100 epoch time **40%**

Neural Scaling Laws: Training Transformers up to 201M Parameters — 10/2025 – 12/2025
- Trained **9 decoder-only Transformer and LSTM** models (**844K – 201M params**) on **100M+ tokens** of structured symbolic sequences, demonstrating **1.2× better scaling** in Transformers with **60%** lower validation loss
- Optimized training on **A100-80GB** using mixed-precision (bfloat16) and Flash Attention, achieving **1.28 test perplexity**

Leadership & Achievements

- **CV Engineer, NYU ARC Robotics (RoboMaster)** – perception & object detection for competitive robotics
- **Harvard CELP Finalist (Top 114 of 10,000+)** | **JEE Advanced Rank: 6111** – Top **0.5% of 1.15M** candidates